AI61201: Visual Computing with AI/ML

Programming Assignment 4: Feature Design (20 Marks)

***Due Date:** **October 27** (by **9 PM** IST)*

***Instructions:** Complete the three tasks given, keeping the following points in mind:*

- *Vectorized implementations are possible for most operations, and using for or while loops to access individual pixels is strongly discouraged due to its inefficiency.*

Task 1: Implement Hough Transform to detect the lines in the given image as follows (1 + 3 + 1 + 1 + 1.5 + 1.5 = 9 Marks):

a. Find the edge map of the image A04_img1.png using Canny edge detector (you can use OpenCV's Canny edge detector directly).
b. Using the detected edge map, find the accumulator matrix (r, theta) for a suitably chosen range of values of r and theta.
c. Visualize the accumulator matrix as a heatmap.
d. Find the peaks in the accumulator matrix using a suitably chosen threshold.
e. Overlay the lines corresponding to the peaks of the accumulator on the input image and display it. Find the length of the longest line in the image.
f. Use OpenCV's HoughLines to detect and show the lines on the given image and display them by overlaying on the input. What is the length of the longest line in the image as per the OpenCV implementation?

Task 2: Land cover classification using GLCM (1+2+2+1+1+1+1+1+1 = 11 Marks)

a. Extract A04_dataset.zip (the zip file contains landcover dataset for land use classification from http://weegee.vision.ucmerced.edu/datasets/landuse.html) Convert the images into grayscale and divide them into training, validation and test sets (70%-15%-15%). Note that since there are 100 images per class, you should perform the train-valid-test split on a per class basis so that the three partitions are balanced.
b. Compute the GLCM matrices from scratch for a chosen number of directions and pixel offsets.
c. For each of the GLCM matrices thus computed, compute the contrast, dissimilarity, homogeneity, ASM, energy, correlation, and entropy features from scratch (the features are as defined in https://scikit-image.org/docs/0.25.x/api/skimage.feature.html#skimage.feature.graycoprops).

d. Use the feature vector of each image from the training set to train a SVM classifier with a suitable kernel (such as RBF). You can directly use the SVM Classifier from scikit-learn for this part.
e. Perform grid search to determine the best number of directions and pixel offsets according to validation set performance obtained with the trained SVM classifier. For grid search, the range for directions to consider is [1, 16] and the range for pixel offsets to consider is [1, 4].
f. For the SVM classifier obtained using the best value of the number of directions and offsets, compute the performance on the test set. Report the performance using accuracy, precision and recall and show the confusion matrix.
g. Use scikit-image's graycomatrix() and graycoprops() to compute the GLCM features of the training set images corresponding to the best values of pixel offsets and directions that you found through cross-validation.
h. Using the features given by scikit-image's implementation of GLCM to train an SVM classifier (with the same hyperparameters such as kernel type etc.).
i. Report the performance of the SVM classifier trained with the GLCM features computed from scikit-image on the test set in terms of accuracy, precision and recall and show the confusion matrix. How does the performance of scikit-image's GLCM features compare with your from scratch implementation?

Link to download the inputs for all tasks

https://drive.google.com/file/d/191d-nm_d7jA_uY30c8X75jjfNXkh9mh5/view?usp=sharing

Submission Guidelines

1. *The content that you submit must be your individual work.*
2. *Submit your code in .py as well as in .ipynb file format. Both these file submissions are required to receive credit for this assignment.*
3. *Ensure your code is well-commented and easy to follow. You can write your answers and explanations using text cells in the jupyter notebook files wherever required. For example, in Task 1 (h) and Task 2 (k), you need to answer using text.*
4. *The files should be named as “<roll_number>_assignment_4”. For example, if your roll number is 23AI91R01, the code the required file names will be 23AI91R01_assignment_4.py and 23AI91R01_assignment_4.ipynb. You should place all these files within a single zip file and upload it to Moodle as 23AI91R01_assignment_4.zip.*

5. *All submissions must be made through Moodle before the deadline. The submission portal will close at the specified time, and submissions via email will NOT be accepted.*
6. *If you have any queries regarding Assignment 4, please email at rajkrishanghosh@gmail.com*